Termeh Taheri

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Profile

ML / Research Engineer specializing in audio + music ML, GPU-accelerated PyTorch experimentation, and scalable model pipelines. Experience with distributed training (DDP/FSDP), profiling, and research prototyping. Built production-grade FastAPI + Docker + AWS systems and integrated text-to-music models into reproducible research workflows.

Experience

AI Engineer, London College of Business Studies Jul 2025 - Present

- Built and deployed production-grade FastAPI + Docker services on AWS EC2 to orchestrate multimodal content-generation workflows around generative models.
- Designed and optimized inference pipelines (batching, caching, async I/O), improving latency and throughput for large-scale LLM-driven generation pipelines.

Software Engineer, Roshan Company Aug 2022 - Aug 2024

- Built and maintained high-performance Python services (Django, FastAPI, Docker) supporting ML-driven products and large-scale data workflows.
- Optimized data pipelines and CI/CD deployments (profiling, batching, caching, indexing), reducing system latency and enabling reproducible releases.

Selected Projects

CMI-Arena — Large-Scale Evaluation Platform for Text-to-Music Models

- Integrated and containerized text-to-music models (Lyria RealTime, DiffRhythm, etc.) as Dockerized FastAPI services for reliable GPU-backed inference.
- Built streaming audio generation pipelines with optimized I/O, batching, and request handling for multi-model evaluation at scale.

SAR-LM — Symbolic Audio Reasoning with Large Language Models (Code)

- Built a modular PyTorch pipeline for audio feature extraction (PANNs, Whisper, MT3, Chordino) and reproducible benchmarking on MMAU/MMAR/OmniBench, enabling large-scale audio reasoning experiments and interpretable model analysis.

Publications

SAR-LM: Symbolic Audio Reasoning with LLMs — Accepted at LLM4MA Workshop @ ISMIR 2025; under review at ICASSP 2026. (Paper)

OmniVideoBench: Towards Audio-Visual Understanding Evaluation for Omni MLLMs — Under review at CVPR 2026. (Paper)

Breast Cancer Prediction by Ensemble Meta-Feature Space Generator Based on Deep Neural Network — Biomedical Signal Processing and Control (2024). (Paper)

Presentation of Encryption Method for RGB Images Based on an Evolutionary Algorithm Using Chaos Functions and Hash Tables — Multimedia Tools and Applications (2023). (Paper)

Education

MSc Artificial Intelligence — Queen Mary University of London (Distinction) September 2025
Thesis: SAR-LM: Symbolic Audio Reasoning with Large Language Models
Advisor: Prof. Emmanouil Benetos

BSc Computer Engineering — Noshirvani University of Technology July 2022
Thesis: Ensemble Deep Learning Algorithm for Medical Image Analysis

Software Proficiency

ML & Deep Learning: Python, PyTorch, TensorFlow, NumPy, SciPy, Scikit-learn, Keras | **Audio & Signal Processing**: Librosa, Essentia, FFmpeg, OpenCV | **Backend & Systems**: FastAPI, Django, Docker, Linux, Celery, Redis, PostgreSQL, ElasticSearch, Pytest | **Cloud & Deployment**: AWS (EC2), CI/CD | **Other**: Java, C++, MATLAB, React, HTML, CSS

Honors & Awards

Chevening Scholarship — Full scholarship recipient for MSc Artificial Intelligence, QMUL 2024–2025
Best Bachelor Project Award, Noshirvani University of Technology — First place across all departments 2022
Research Grant (No P/M/1110) — Supported publication in *Biomedical Signal Processing and Control* 2021